Abstract

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Abstract

Multi-agent active inference gives a natural account of belief sharing: agents hold local posteriors over a shared latent state, communicate those beliefs, and pool them into a colony-level consensus. The same mechanism is fragile when a member is miscalibrated, corrupted, or strategically wrong.

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Abstract (part 2)

Because the standard pool multiplies the reports together, a single confident-but-wrong broadcast that puts near-zero mass on the true state can pull the whole consensus off it, outweighing many honest members.

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Abstract (part 3)

The colony therefore needs a way to preserve the useful structure of belief sharing while limiting the influence of contaminated beliefs.

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Abstract (part 4)

This paper presents Active Fedference, a discrete-categorical framework that connects robust federated generalized variational inference with active inference belief sharing. The main bridge is structural: standard belief sharing appears as the non-robust corner of a broader generalized-Bayes family,

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Abstract (part 5)

while robust losses, conservative server fusion, and explicit aggregation diagnostics describe how the system moves away from that corner under declared contamination mechanisms.

Abstract (part 6)

The result is not a replacement for belief sharing, but a containment result: ordinary belief sharing is recovered when robustness is turned off. Bounded-loss theory applies on the client axis, while the variational-server axis supplies an objective-backed redescending weight update.

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Abstract (part 7)

Following the manuscript's authoritative three-axis guarantee map, the paper keeps source-conditional client updates, a recovery-only heuristic server rule, and an objective-backed variational server rule as separate claim owners, with no robustness guarantee transferred between them.

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Abstract (part 8)

The study suite then exercises the framework as an end-to-end research system: recovery checks anchor the standard-Bayes limit, belief-sharing studies verify the communication baseline, contamination experiments test robust consensus, and extension studies probe moving agents, hierarchical latent structure,

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Abstract (part 9)

sensitivity to acuity and colony size, parameter recovery, and single-host socket-backed federation traces.

Abstract (part 10)

All reported quantities are generated from deterministic analysis artifacts and injected into the manuscript by token, so the paper, figures, release package, and validation reports remain tied to the same execution record.

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Abstract (part 11)

The open-source repository is ActiveInferenceInstitute/Active_Fedference. This final-release manuscript uses assigned version DOI 10.5281/zenodo.22149133; the bound deposited PDF and repository release are controlled by separate publication gates.

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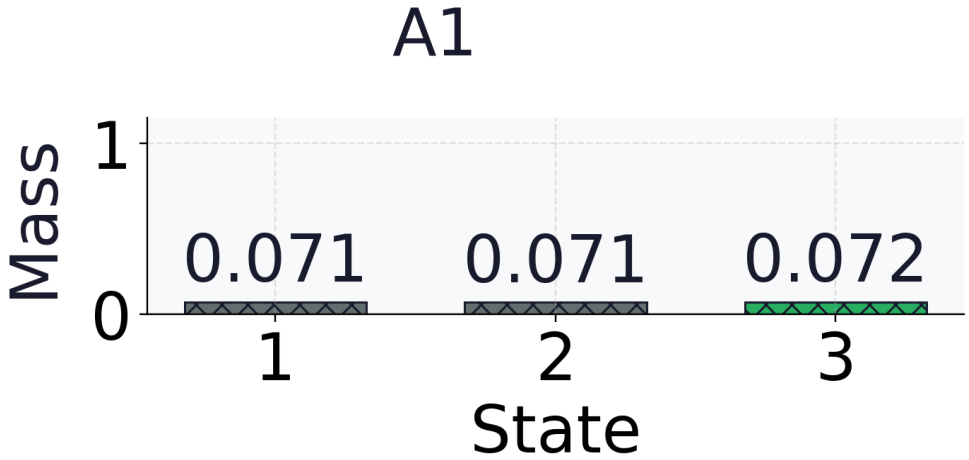
Abstract (part 12)

Keywords: active inference, federated learning, generalised variational inference, belief sharing, robustness, FedGVI

The complete system schematic is shown in fig. 4.

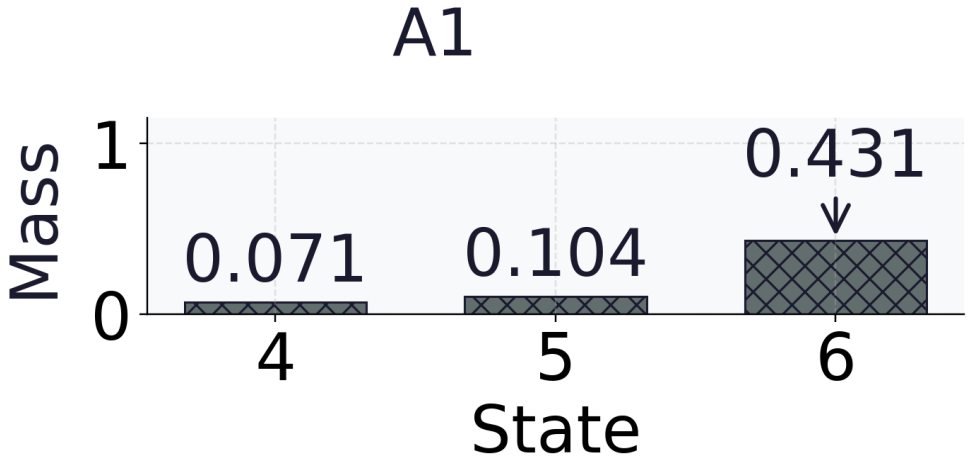
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Abstract (part 13)

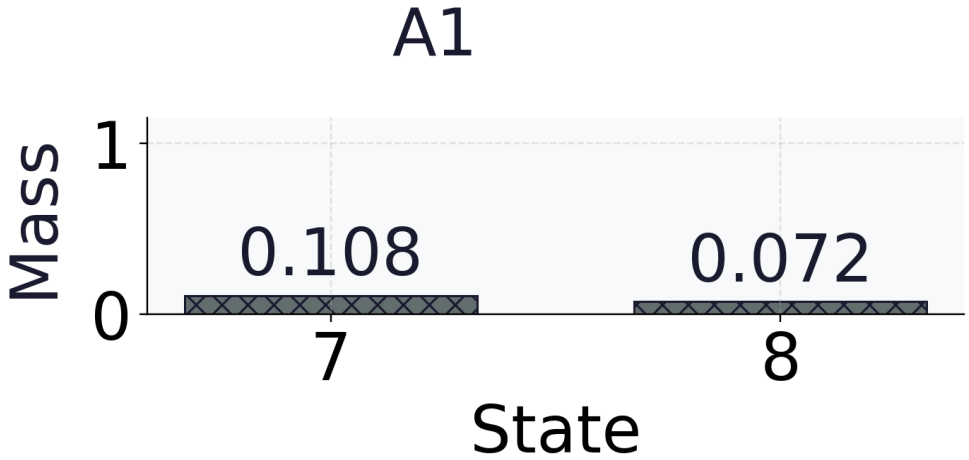


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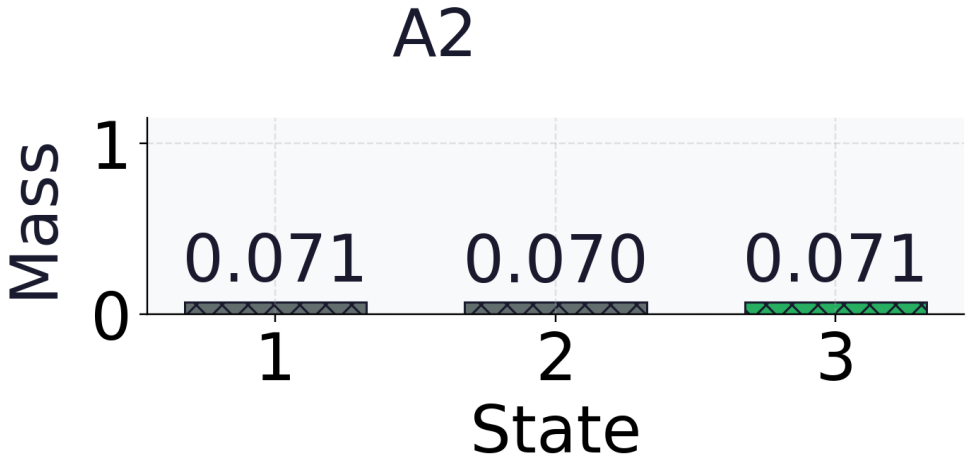
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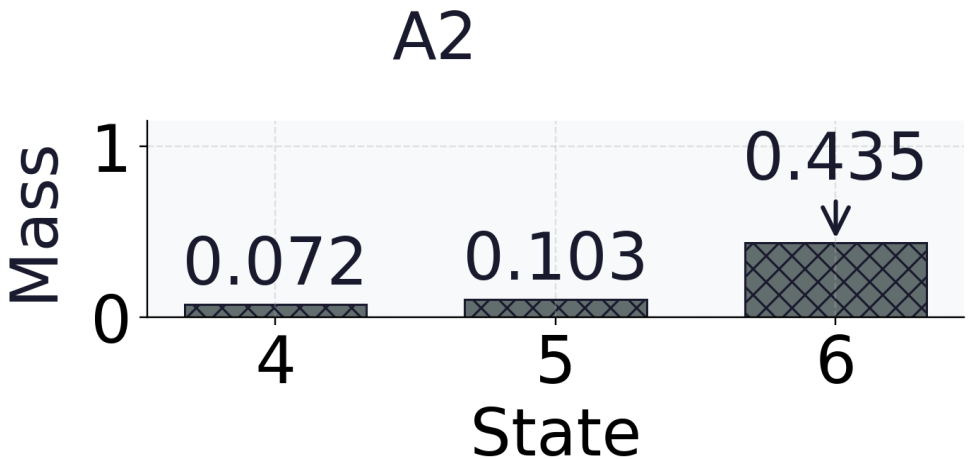
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Abstract (part 16)

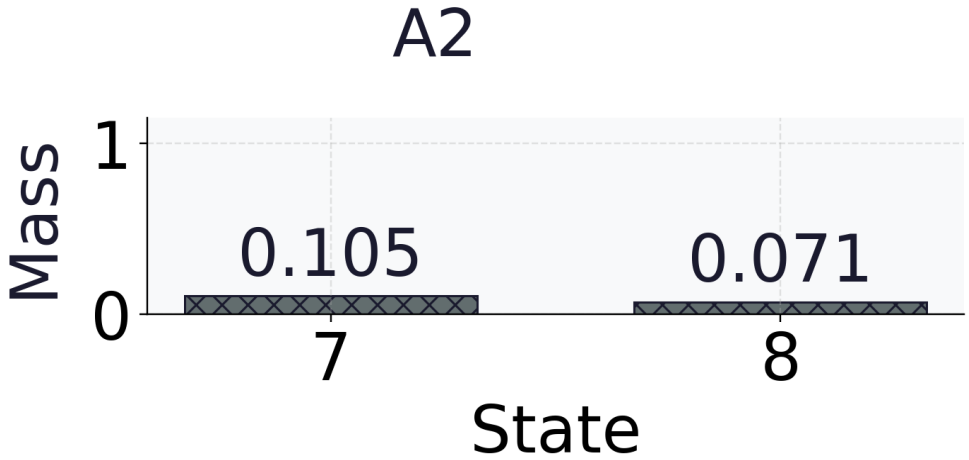


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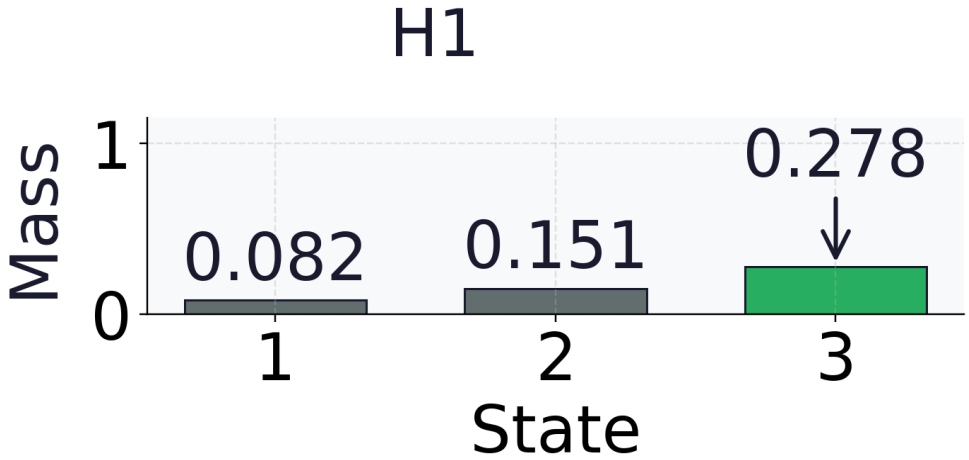


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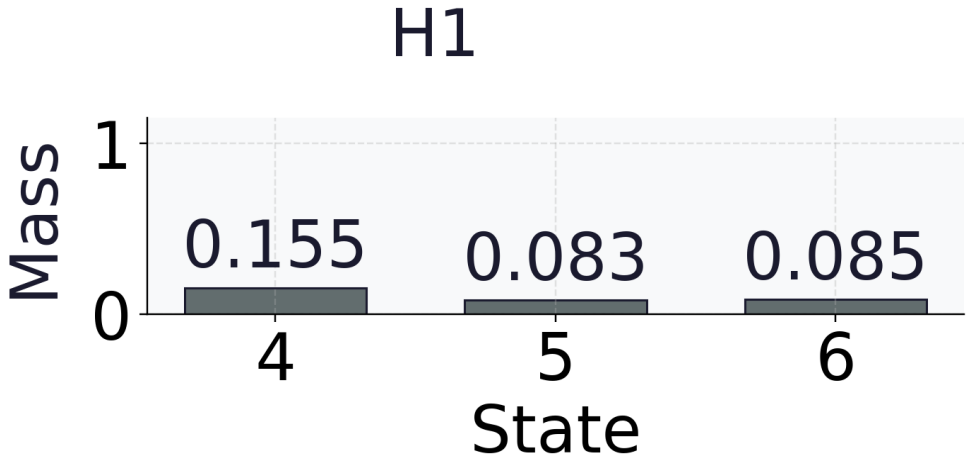
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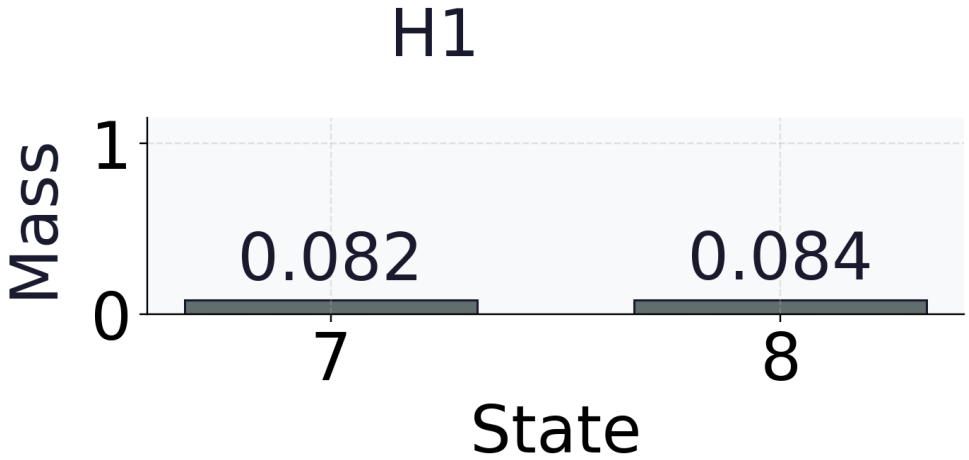
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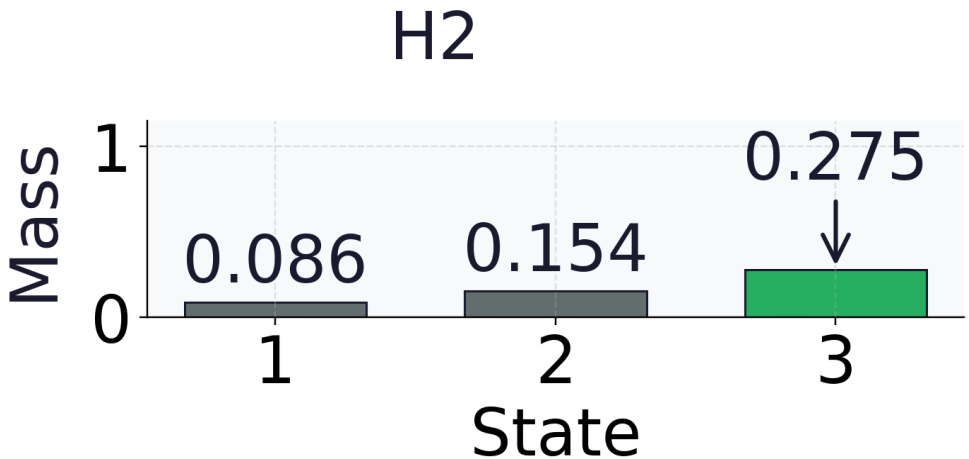
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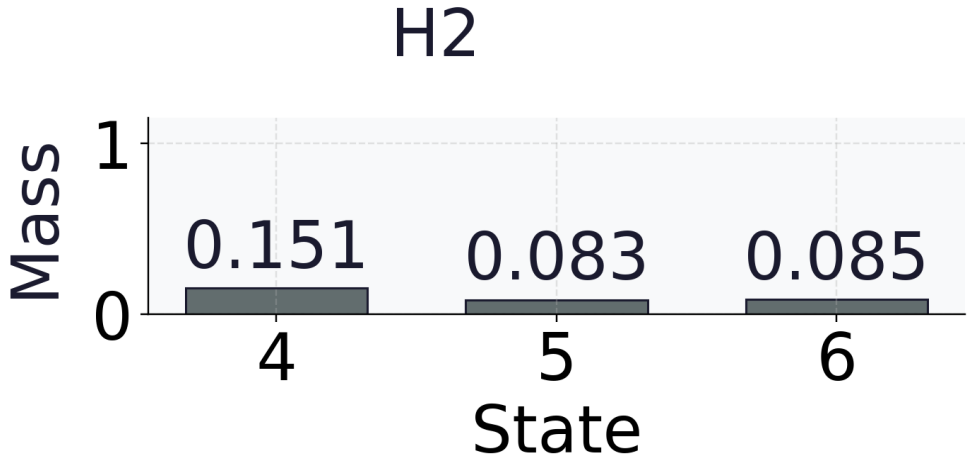
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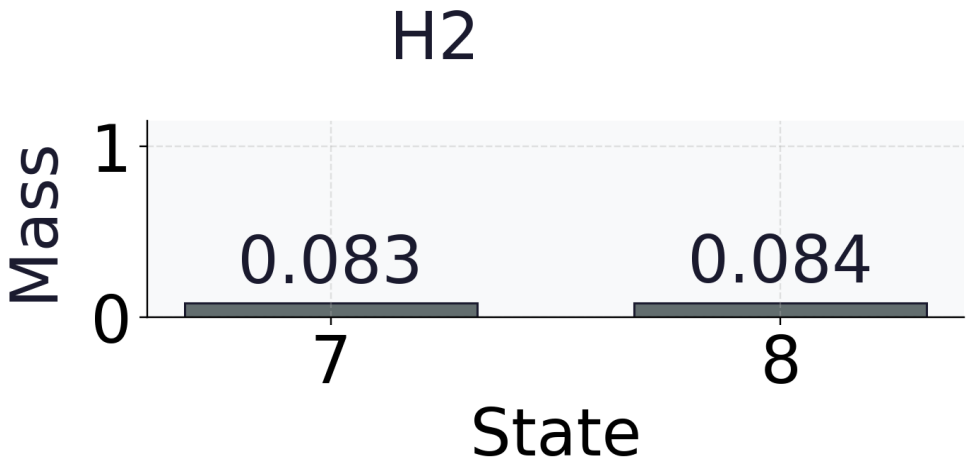


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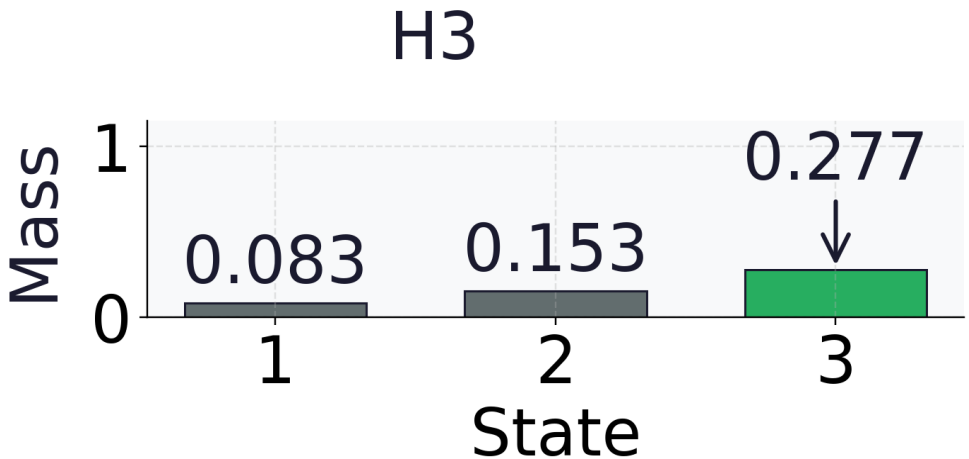


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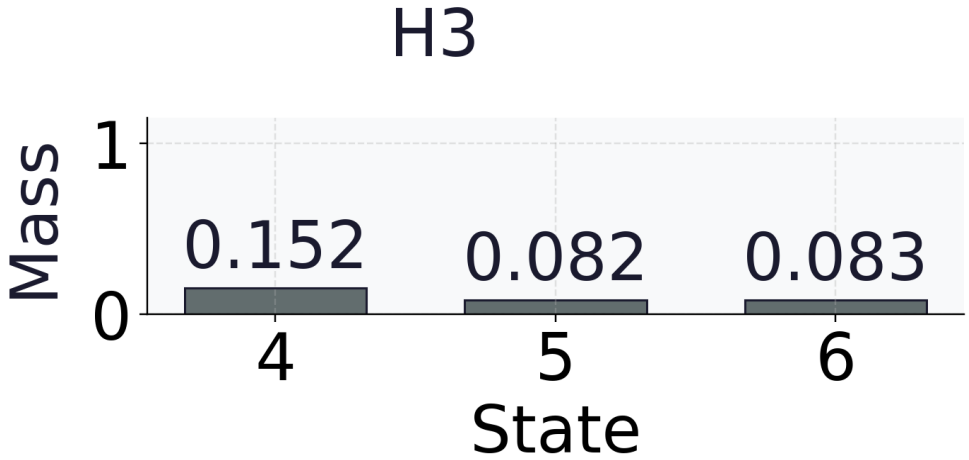
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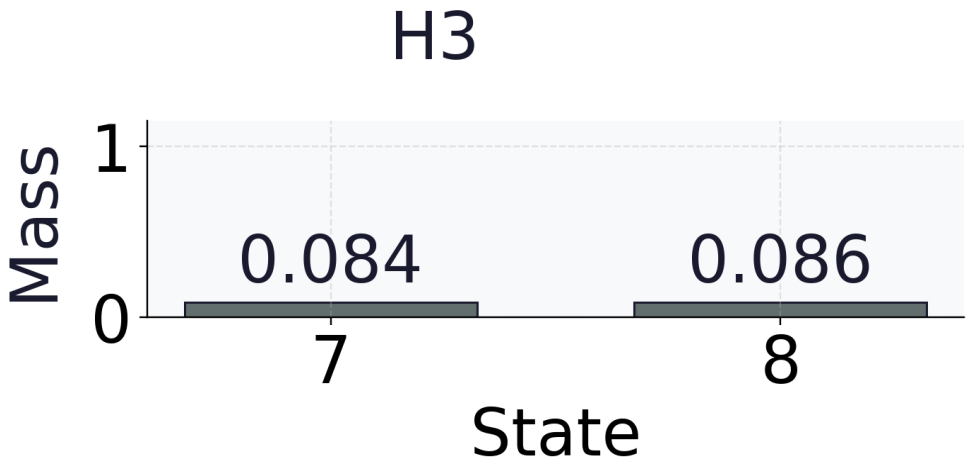


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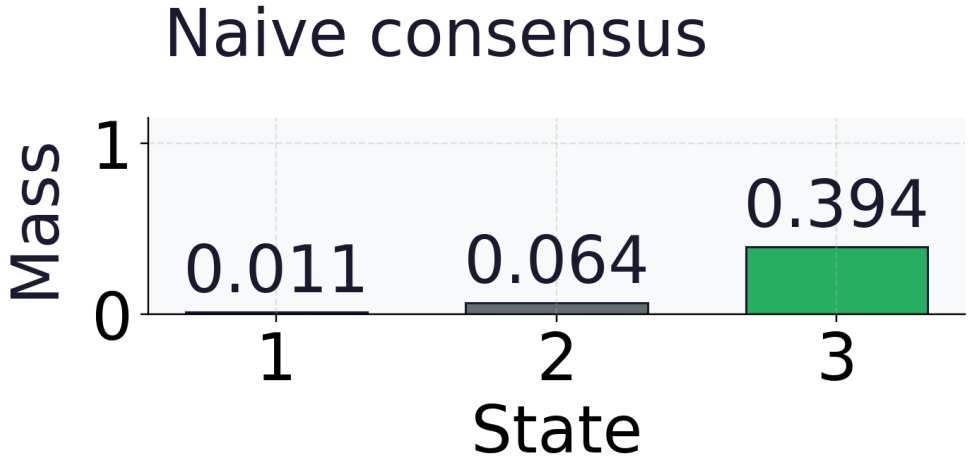


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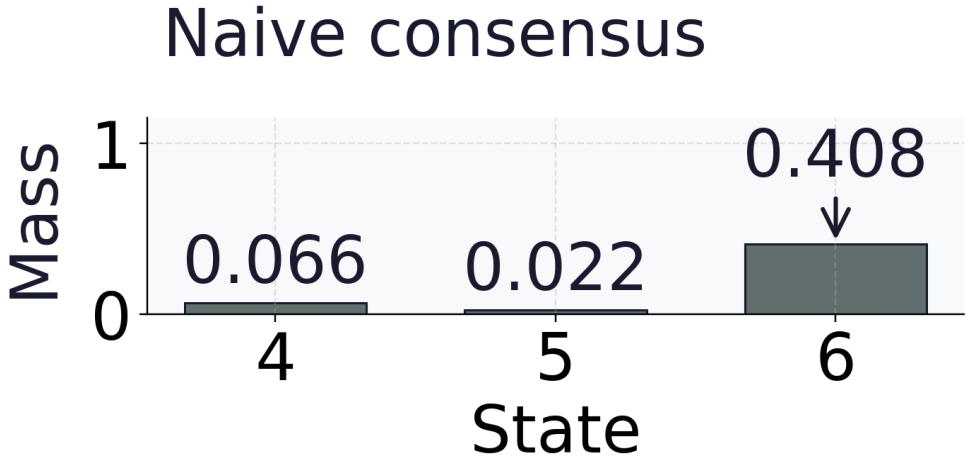


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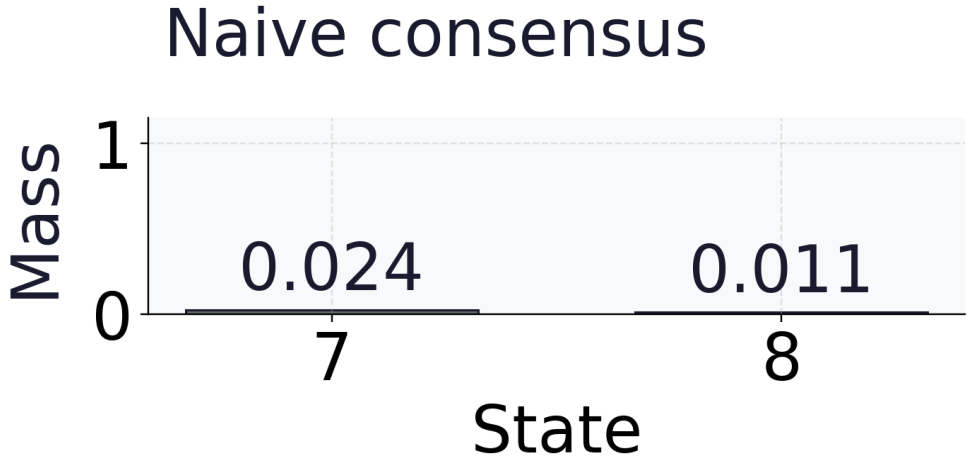
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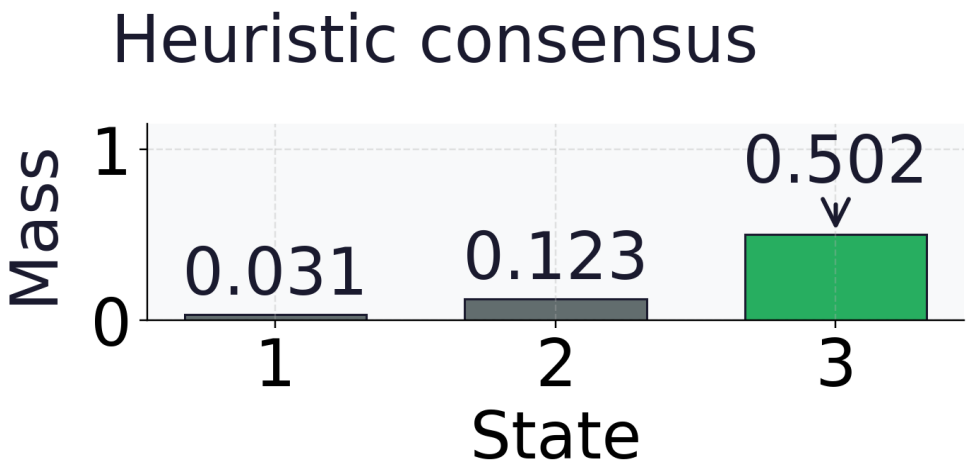


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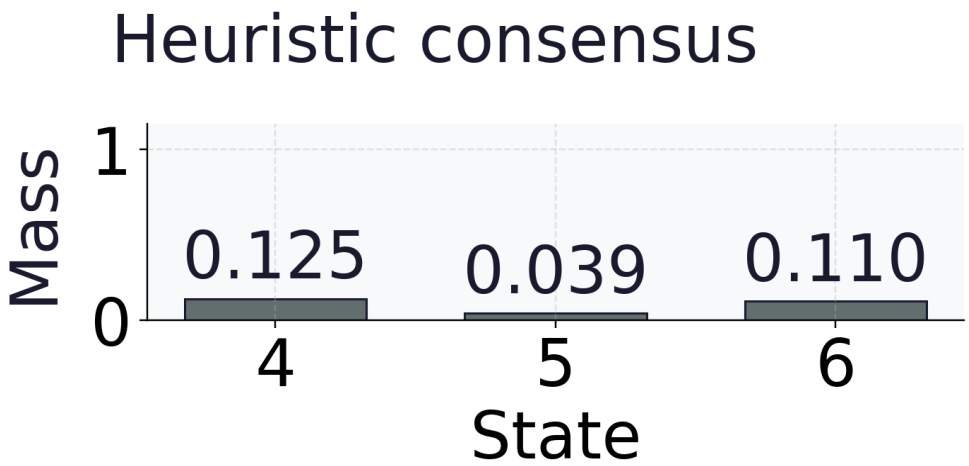


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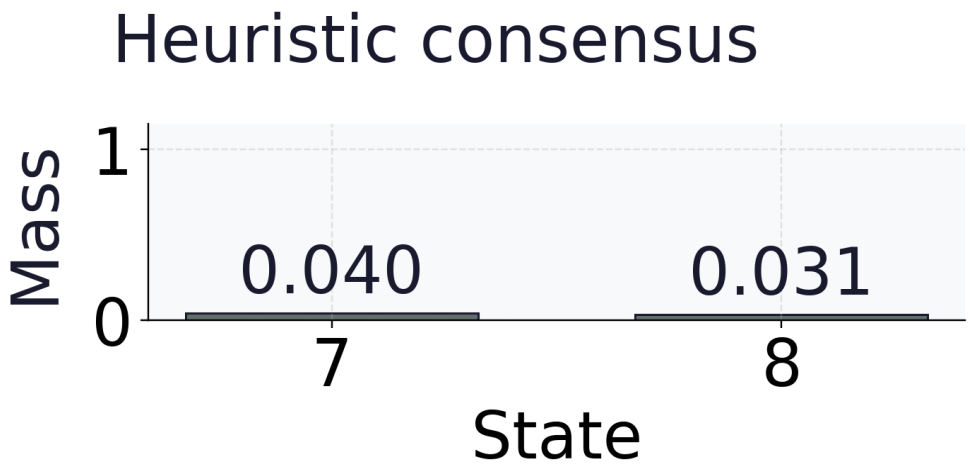
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Abstract (part 32)



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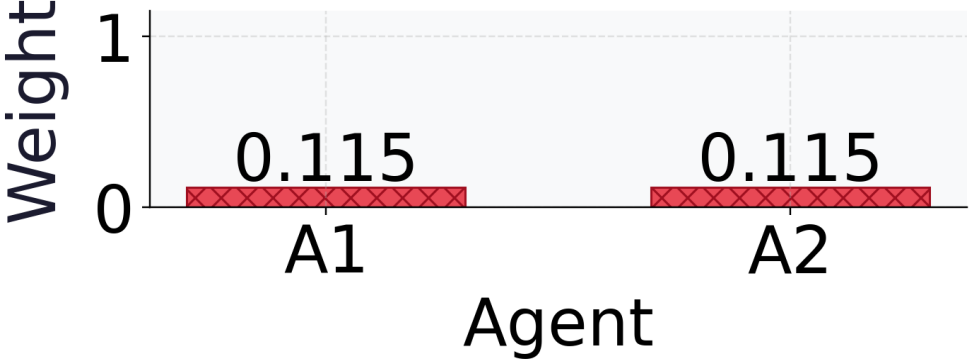
Abstract (part 33)



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Abstract (part 34)

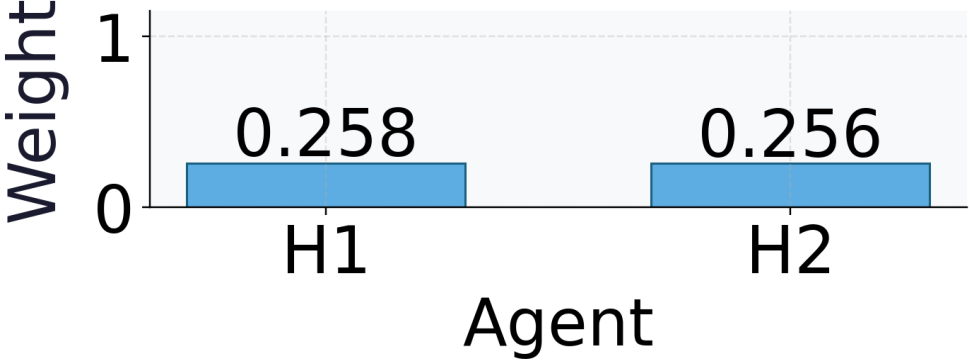
Effective influence weights



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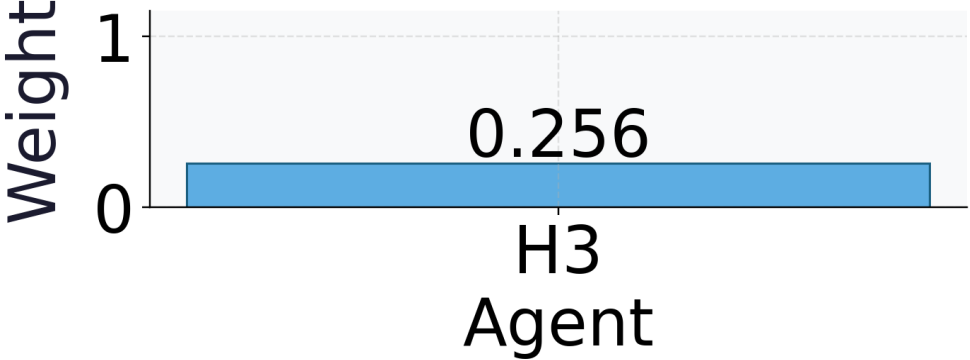
Abstract (part 35)

Effective influence weights



Abstract (part 36)

Effective influence weights



Abstract (part 37)

Configured colony: 5 agents; 2 adversarial, 3 honest. Contamination: 40%. True state: 3.

Abstract (part 38)

Naive equal-weight log pool:
true-state mass 39%; argmax state 6.

Abstract (part 39)

Heuristic robust aggregation:
true-state mass 50%; argmax state 3.
Robustness 1.5.

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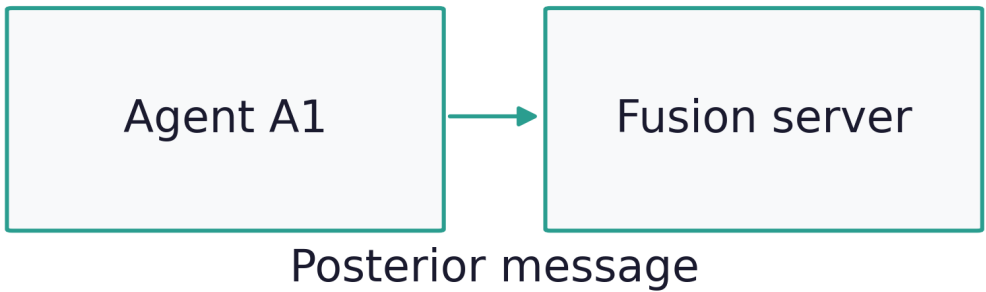
Abstract (part 40)

A / crosshatching: adversarial; H / plain: honest. All probability panels share a 0–1 scale. The true state is green; other states are gray.

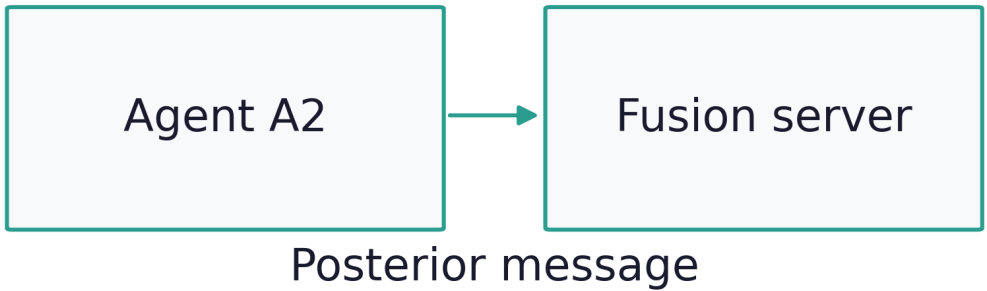
Abstract (part 41)

One deterministic configured colony, not a replicated benchmark. True-state mass percentages are not deployment accuracy estimates. No uncertainty interval applies.

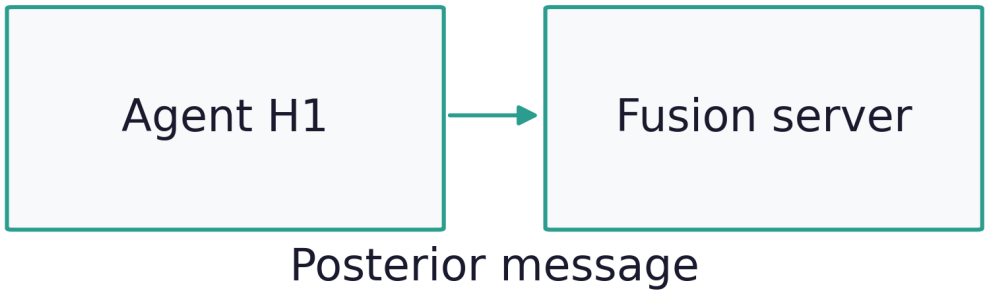
Abstract (part 42)



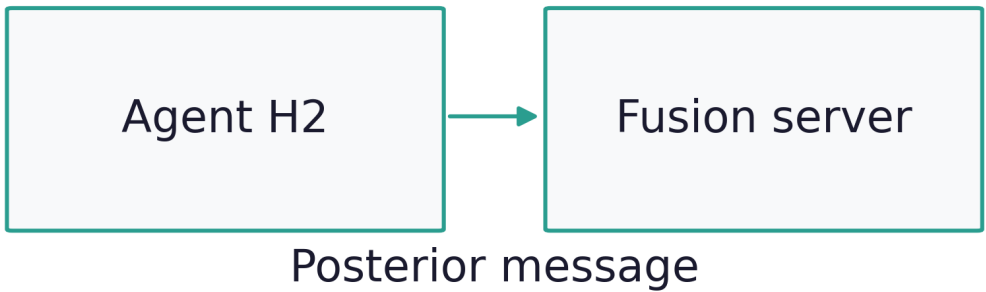
Abstract (part 43)



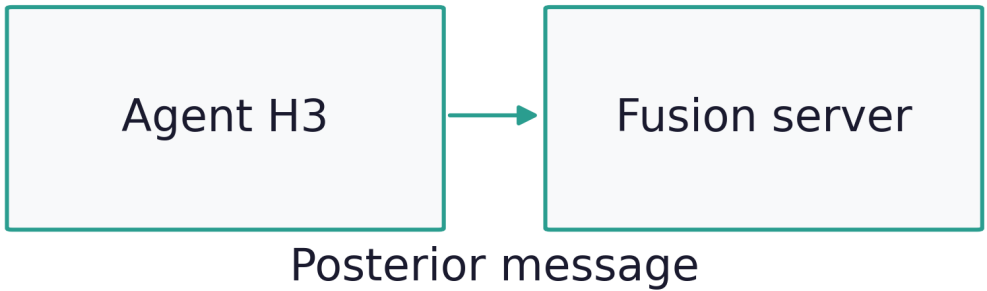
Abstract (part 44)



Abstract (part 45)



Abstract (part 46)



Abstract (part 47)

The variational server is a separate objective-backed route. The displayed naive-versus-heuristic colony values do not measure its performance.